7/8/26

## Complexity Table

| $\log n$ | $n$ | $n\log n$ <br> $\|\|$ <br> $\log n!$ | $n^2$ | $n^3$ | ... | $2^n$ | $n!$ | $n^n$ |
|---|---|---|---|---|---|---|---|---|

**To show:** $\log n! \in \Theta(n\log n)$.

$$\log n! = \log\left(n \cdot (n-1) \cdots 2 \cdot 1\right)$$

Revision ↓

$$= \log n + \log(n-1) + \cdots + \log 2 + \log 1$$

$$= (\log n + \log 1) + (\log(n-1) + \log 2)$$

$$+ \left(\log\left(\frac{n}{2}\right) + \log\left(\frac{n}{2} - 1\right)\right)$$

$$> \log \frac{n}{2} + \log \frac{n}{2} + \cdots + \log \frac{n}{2}$$

$$= \frac{n}{2} \log \frac{n}{2}$$

$$\log n! > \frac{n}{2}\log\frac{n}{2}$$

$$= \left(\frac{n}{2}\log n\right) - \left(\frac{n}{2}\log 2\right)$$

$$\geq \left(\frac{n}{4}\log n + \frac{n}{4}\log n\right) - \frac{n}{2}\log 2.$$

$$= \frac{n}{4}\log n + \left(\frac{n}{4}\log n - \frac{n}{2}\log 2\right)$$

For what values of $n$ does this dominate?

$$= \frac{n}{4}\log n + n\left(\frac{1}{4}\log n - \frac{1}{2}\log 2\right)$$

$$\geq \frac{n}{4}\log n \quad \text{for } n \geq 4.$$

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After what $n$, does $\left(\frac{1}{4}\log n - \frac{1}{2}\log 2\right)$ dominate?

ie $\frac{1}{4}\log n - \frac{1}{2}\log 2 \geq 0$.

$$\frac{1}{4}\log n \geq \frac{1}{2}\log 2.$$

$$\log n \geq 2\log 2 = \log 4$$

Choose $n \geq 4$

Take $n_0 = 4$.

$\Rightarrow \log n! \geq \frac{n}{4}\log n$ for $n \geq 4$.

$\Rightarrow \log n! \in \Omega(n \log n)$ — ①

Choose $c = \frac{1}{4}$

$n_0 = 4$.

To show: $\log n! \in O(n \log n)$

$$n! = n \cdot (n-1)(n-2) \cdots 2 \cdot 1$$

$$\leq n \cdot n \cdot n \cdots n$$

$$= n^n.$$

$$\log n! \leq \log(n^n) = n\log n$$

$$\Rightarrow \log n! \in O(n\log n) \quad -(2)$$

Hence, by (1) and (2),

$$\log n! \in \Theta(n\log n)$$

Bubblesort

↓

Selection Sort

| | Comparisons | Swaps |
|---|---|---|
| Bubblesort | $n^2$ | $n^2$ |
| Selection Sort | $n^2$ | $n-1$ |

# Selection Sort:

| | | |
|---|---|---|
| pos=1 | 80 | → Input. |
| pos=1 | 14 | |
| pos=3 | 90 | |
| pos=3 | 30 | |
| pos=3 | 25 | |
| pos=3 | 45 | |

pos=3 → position of maximum number.

Swap value at pos=3 ⟷ Value at last pos.

| Ist pass | | 2nd Pass | | 3rd Pass | | 4th Pass | | 5th Pass | 6th Pass |
|---|---|---|---|---|---|---|---|---|---|
| pos=1 | 80 | pos=1 | 80 | pos=1 | 25 | pos=1 | 25 | 25 | 14 |
| pos=1 | 14 | pos=1 | 14 | pos=1 (25>14) | 14 | pos=1 | 14 | 14 | 25 |
| pos=3 | 90 | pos=1 | 45 | pos=3 (45>25) | 45 | pos=3 | 30 | 30 | 30 |
| pos=3 | 30 | pos=1 | 30 | pos=3 (45>30) | 30 | | 45 | 45 | 45 |
| pos=3 | 25 | pos=1 | 25 | | 80 | | 80 | 80 | 80 |
| pos=3 | 45 | | 90 | | 90 | | 90 | 90 | 90 |

Worst Case:

Total Swaps:

- One swap in each column (pass)
- Total no of passes = $n-1$.

$\therefore$ Total Swaps = $n-1$.

Q: $n-1 \in \Theta(n)$

$n-1 \leq n \Rightarrow n-1 \in O(n)$

Want: $n-1 \geq cn$ for some constant $c$

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What is the $c$ that makes this after some $n$? happen.

Take $c = \frac{1}{2}$.

| | $n-1$ | $n/2$ |
|---|---|---|
| $n=1$ | 0 | $1/2$ |
| 2 | 1 | 1 |
| 3 | 2 | 1.5 |
| 4 | 3 | 2 |
| | ⋮ | ⋮ |

Hence

$n-1 \geq cn/2$

for all $n \geq 2$

and $c = \frac{1}{2}$.

$\Rightarrow n-1 \in \Omega(n)$

Another way:

$$\lim_{n\to\infty} \frac{n-1}{n} = \lim_{n\to\infty} 1 - \frac{1}{n}$$

$$= 1 \neq 0.$$

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$ then $f(n) < g(n)$

$$\Rightarrow f(n) \in O(g(n))$$

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = k \neq 0 \Rightarrow f(n) \in \Theta(g(n))$

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$ $\Rightarrow f(n) > g(n) \Rightarrow f(n) \in \Omega(g(n))$.

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# Selection Sort:

Input: $A[1, \ldots, n]$

For $i \leftarrow 1$ to $n-1$

  $pos \leftarrow 1$

  For $j \leftarrow 1$ to $n-i$

    If $A[j] > A[pos]$ then $pos \leftarrow j$

  $A[pos] \longleftrightarrow A[n-i+1]$

Variable:

$i \leftarrow$ running through column.

$j \leftarrow$ running through elmts of each column.

| i | last | |
|---|---|---|
| 1 | $n$ | $n-1+1$ |
| 2 | $n-1$ | $n-2+1$ |
| 3 | $n-2$ | $n-3+1$ |

$n-(i-1)$
$= n-i+1$

double sum
↗ ↘
two for loops

## Number of Comparisons:

$$T(n) = \sum_{i=1}^{n} \sum_{j=1}^{n-i} 1$$ ↗ 1 comparision.

$$= \sum_{i=1}^{n} n-i$$

$$= (n-1) + (n-2) + (n-3) + \cdots + 1$$

$$= \frac{n(n-1)}{2} \in \Theta(n^2)$$

Number of swaps :

$$S(n) = \sum_{i=1}^{n-1} 1 = n-1 \in \Theta(n).$$

**Result:** Finding max (& min) in an array of size $n$ takes $n-1$ comparisions.

## Code to find max:

Input: $A[1, 2, \cdots, n]$

For $i \leftarrow 2$ to $n-1$

pos $\leftarrow 1$

If $A[pos] > A[i]$ then pos $\leftarrow i$

print $A[pos]$

$$T(n) = \sum_{i=2}^{n-1} 1 = n-1 \in \Theta(n)$$

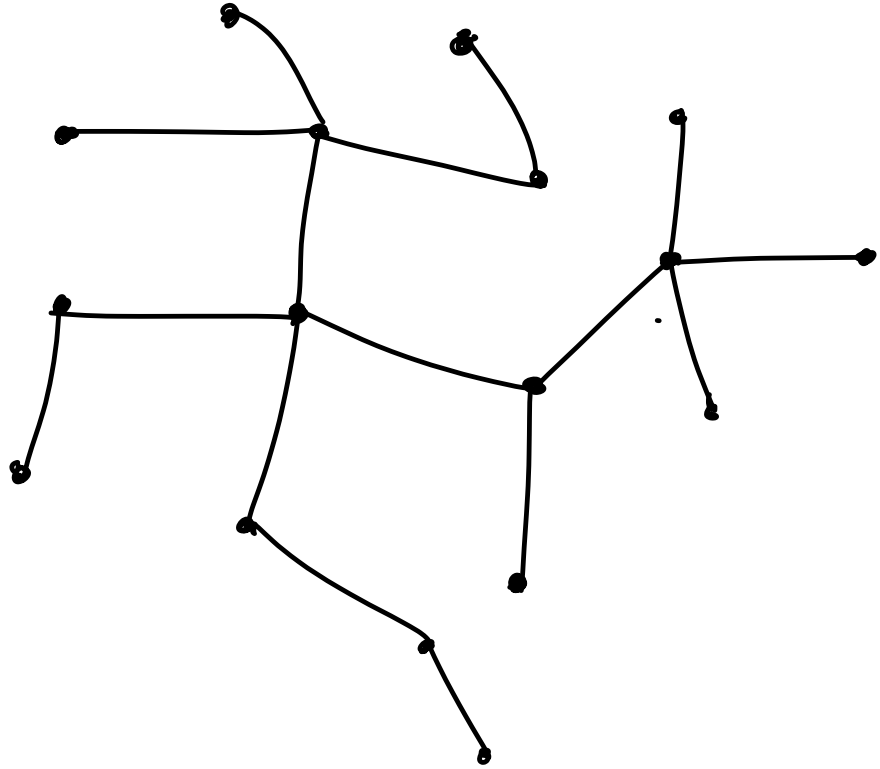
Claim: This code is optimal. / This code cannot be improved from $O(n)$

**Result**: Finding max (& min) in an array of size $n$ takes $n-1$ comparisions.

**Proof of Result**: (Proof by contradiction)

If possible there exists an algorithm A which takes $< n-1$ comparisions.

**To Prove**: A is incorrect.



n=17
16 edges.

A tree · n vertices, n-1 edges.

↓

Connected graph with no cycles. "minimally connected graph".

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(Constructing the graph for the algorithm)

If the algorithm A compares A[i] with A[j] then connect A[i] with A[j].

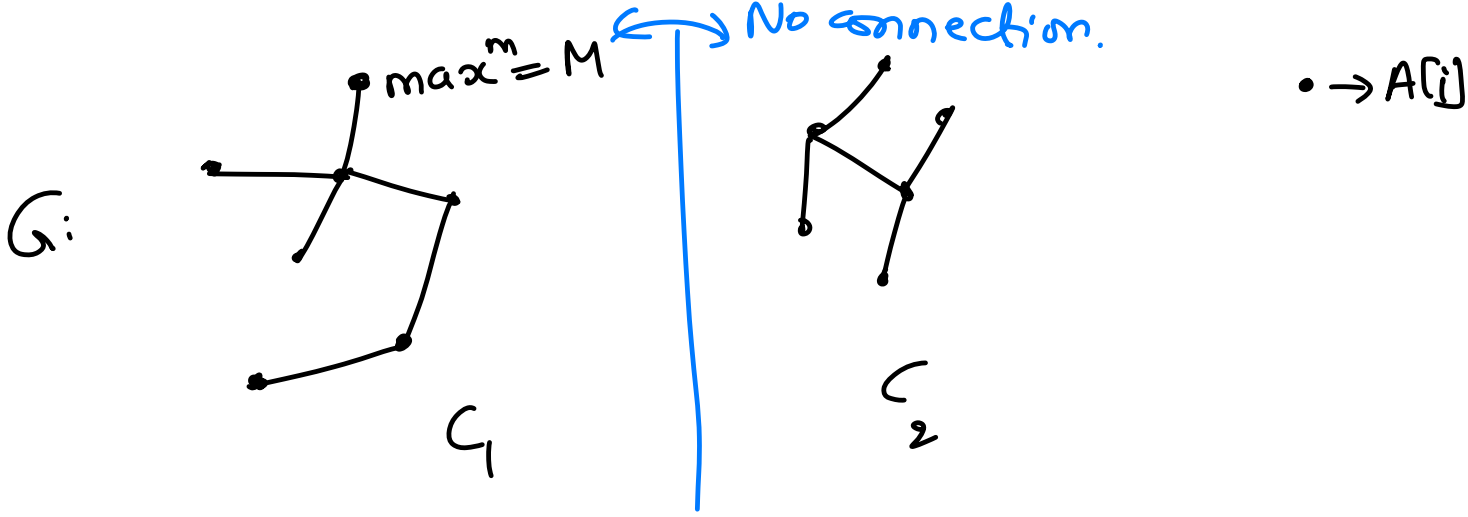
Q: What happens if there are $< n-1$ comp.

The structure is not a tree because it will be **disconnected** graph

$\Rightarrow$ Since there are $< n-1$ comparisions, the corresponding graph is not a tree.

$\Rightarrow$ There will be atleast 2 disconnected components for the graph $C_1$ and $C_2$.

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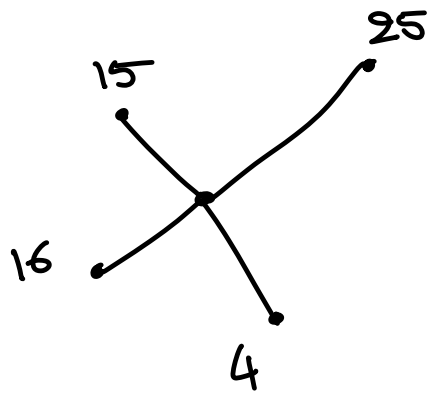

(The max A[pos] must be in one of two components)

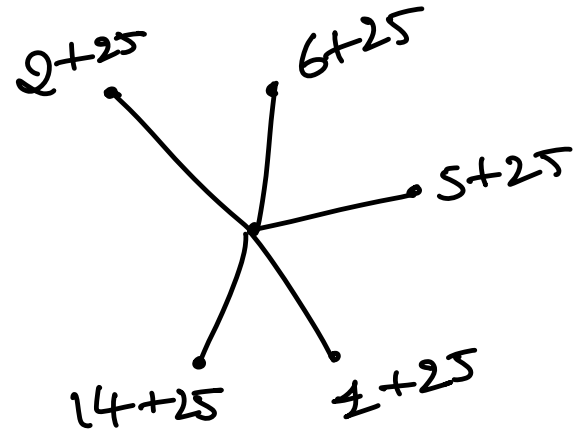
Suppose without loss of generality, say max$^{m}$ value is for $C_1$

Idea: Give an input s.t this algorithm fails.

Trick: Add M to each vertex value of $C_2$ and give this as input to A.

eg:




Alg. A still gives the output as M because A is not doing any comparisions with elements of $C_2$.

$\Rightarrow$ Alg. A is incorrect.

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$\Rightarrow$ finding max/min algorithm is $\Theta(n)$.

(Tight Algorithm)